

InsightIQ: Mathematical Framework for Business Optimization

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1. Project Overview

In today's data-driven business environment, organizations generate large volumes of sales, inventory, supplier, and customer data. Although Business Intelligence (BI) systems provide descriptive analytics and visual reports, they often lack the capability to make mathematically optimized business decisions. Critical operations such as supplier selection, demand forecasting, inventory planning, and product pricing require rigorous mathematical analysis to balance multiple constraints while maximizing profitability and operational efficiency. This project proposes a Mathematical Decision Engine that integrates predictive modelling and optimization techniques to provide intelligent business recommendations. The project demonstrates the practical application of mathematical concepts from Mathematical Foundations for Computing (MFC) in solving complex business optimization problems.

2. Problem Statement / Research Gap

Most existing Business Intelligence systems primarily analyse historical business data and present it through dashboards and reports. While these systems support descriptive analysis, they do not mathematically determine the optimal supplier, forecast future demand with uncertainty, or recommend profit-maximizing pricing strategies under multiple constraints. Existing approaches often address supplier optimization, demand prediction, and pricing as independent problems, resulting in fragmented decision-making and reduced business efficiency.

This project addresses this gap by developing a unified mathematical decision engine that integrates demand forecasting, probabilistic modelling, constrained optimization, and pricing optimization into a single analytical framework. The proposed system enables organizations to make scientifically justified decisions by simultaneously considering multiple business objectives and operational constraints.

3. Objectives

The primary objectives of the proposed project are:

- To design a mathematical decision engine for intelligent business decision-making.
- To forecast future product demand using Weighted Least Squares (WLS) based on historical sales patterns.
- To model customer demand transitions and market uncertainty using Markov Chains.
- To formulate supplier selection as a Constrained Convex Optimization problem while satisfying procurement cost, supplier capacity, inventory requirements, delivery schedules, and budget constraints.
- To verify the optimality of the obtained solutions using Karush-Kuhn-Tucker (KKT) Conditions.
- To determine optimal product pricing through Gradient-Based Optimization in order to maximize expected profit.
- To integrate these mathematical models into a unified framework capable of generating optimized business recommendations.

4. Mathematical Models and Optimization Techniques

- **Weighted Least Squares (WLS):** Used to estimate future product demand by assigning appropriate weights to historical observations, thereby improving forecasting accuracy.
- **Markov Chains:** Applied to model probabilistic transitions between demand states, enabling prediction of future customer demand behaviour under uncertainty.
- **Constrained Convex Optimization:** Formulates supplier selection as an optimization problem that minimizes procurement cost while satisfying inventory, supplier capacity, delivery time, and budget constraints.
- **Karush-Kuhn-Tucker (KKT) Conditions:** Used to verify that the constrained optimization problem satisfies the necessary optimality conditions and that the obtained solution is mathematically valid.
- **Gradient-Based Optimization:** Optimizes the selling price by iteratively maximizing the profit function while considering predicted demand and procurement cost within a convex optimization framework.